

assignment 11 - stellar parameters I

 This is a preview of the published version of the quiz

Started: Mar 22 at 7:36pm

Quiz Instructions

2 attempts . I am available for questions after our class. Use the slides spectral parameters unit 7.1

you will use the sim:

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html 
(https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html)



Question 1 3.5 pts

Stars more massive than our Sun

☐

mass and life are not correlated

☐

There is no star larger than our Sun

☐

live a longer life

☐

live a shorter life



Question 2 3.5 pts

The absorption lines of a star moving toward us (as seen from Earth) , will be

☐

redshifted

☐

there is no line in a spectrum

☐

thicker

☐

blueshifted



Question 3 3.5 pts

A star 10 times more massive than the Sun will have a luminosity (energy output per unit time or power)

- See slide with relationship L and M (luminosity and mass)- unit 7.1



x 100



x 10,000



x 10



I miss the class



x 1000



Question 4 3.5 pts

Jupiter and the Sun orbit their common center of mass. what is True

hint: for every action, there is an equal an opposite reaction.



The sun does not move at all.



The sun wobbles under the influence of Jupiter



Question 5 3.5 pts

Annie-Jump Cannon came up with the classification: O, B, A, F, G, K, M

Then the following classes were added



T, L, H



L, T, Y



T, J, S



L, H, S



Question 6 3.5 pts

A star 5 times more massive than our sun will live for about:

hint: the Sun lives for 10 billion years or 10,000,000,000 years (life is inversely proportional to the Mass raised to the third power. ($5 \times 5 \times 5 = 125$). See slide 5 (unit 7.1)



400 million years (400,000,000)

☐

1 million years

☐

80 million years

☐

3 million years



Question 7 4.5 pts

Which of the following stars has the hottest surface temperature?

☐

the yellow star

☐

the blue star

☐

the red star

☐

the green star

☐

the orange star



Question 8 3.5 pts

What is a star's *luminosity*? unit 7.1

☐

the surface temperature of the star

☐

the apparent brightness of the star in our sky

☐

the total energy that the star radiates each second

☐

the lifetime of the star



Question 9 3.5 pts

A little green thing live only on chocolate cookies. it has 100 cookies and eat 25 cookies per minute. How long it is expected to live ?

☐

4 minutes

☐

25 minutes

☐

100 minutes

☐

5 minutes



Question 10 3.5 pts

The black body curve was introduced in 1901 by .Max PLank. IT shows that the amount of radiation emitted by a star and its color (or hot object) only depends on

☐

its composition

☐

its size

☐

its distance from us

☐

its temperature



Question 11 3.5 pts

Consider a star 5 times more massive than the Sun. It will be _____ times more luminous than the Sun. (slide. luminosity is proportional to the mass to the 4th power)

☐

25

☐

125

☐

5

☐

625



Question 12 3.5 pts

For main-sequence stars, which of the following is *true* about low-mass stars compared to high-mass stars?

☐

Low-mass stars are cooler and less luminous than high-mass stars.

☐

Low-mass stars are hotter and more luminous than high-mass stars.

☐

Low-mass stars have the same temperature and luminosity as high-mass stars.

☐

Low-mass stars are cooler but more luminous than high-mass stars.

☐

Low-mass stars are hotter but less luminous than high-mass stars.



Question 13 3.5 pts

Which of the following statements about spectral types of stars is always *true*?



A star with spectral type G has a lower interior temperature than a star with spectral type B.



A star with spectral type G has a lower mass than a star with spectral type B.



All of the above are true.



A star with spectral type G has a cooler surface temperature than a star with spectral type B.



A star with spectral type G has a smaller radius than a star with spectral type B.



Question 14 3.5 pts

Which of the following best describes the axes of a Hertzsprung-Russell (H-R) diagram?



mass on the horizontal axis and stellar age on the vertical axis



mass on the horizontal axis and luminosity on the vertical axis



surface temperature on the horizontal axis and radius on the vertical axis



surface temperature on the horizontal axis and luminosity on the vertical axis



interior temperature on the horizontal axis and mass on the vertical axis



Question 15 3.5 pts

Which of the following people reorganized the spectral classification scheme into the one we use today and personally classified over 400,000 stars?



Annie Jump Cannon



Henry Draper



Williamina Fleming



Cecilia Payne-Gaposchkin



Edward Pickering



Question 16 3.5 pts

All the stellar parameters depend on the mass except



energy output



size



life expectancy



chemical composition



Question 17 3.5 pts

Which stellar spectral type has the strongest *molecular* absorption lines (molecules getting excited to higher level of energy) in its visible light spectrum? (unit 7.1 slide 19).



G



K



B



M



O



Question 18 3.5 pts

Which spectral type represents stars with the hottest surface temperatures?



G



A



B



M



O



Question 19 3.5 pts

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html

[\(https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html\)](https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html)

check everything you can check. Move the cursor to Sirius. On the x-axis you can see the peak wavelength (its in yellow). the star gives out the most radiation in this color. What is this peak wavelength? (using this value you can find the temperature of Sirius. Cool, no?)

☐

about 0.1 μm

☐

about 0.5 μm

☐

about 1 μm

☐

about 0.3 μm



Question 20 3.5 pts

If you spend 2 gallon per day (driving), how many days you can last with 10 gallons?

☐

I miss the class

☐

2 days

☐

5 days

☐

10 days



Question 21 3.5 pts

According to the quote in the famous movie blade runner. A light that shines twice as much, lives (true for stars)

☐

half as long

☐

8 times less

☐

4 times less

☐

twice as long



Question 22 3.5 pts

Use:

https://phet.colorado.edu/sims/html/blackbody-spectrum/latest/blackbody-spectrum_en.html

check everything you can check. The peak wavelength of the Sun is about (μm is micro meter , one millionth)

☐

0.5um

☐

0.1 um

☐

50 um

☐

5 um



Question 23 3.5 pts

building on number 24, Multiply the temperature of the Sun by its peak wavelength.

Then select sirius and multiply its peak wavelength by its temperature. What are the 2 numbers you get?

☐

2900, 3500

☐

2900,1500

☐

2900, 2900

☐

5800,10000



Question 24 3.5 pts

PLay with the simulation. As you increase the temperature, the area under the curve (energy per unit second output)

☐

increases

☐

stays the same

☐

No clue

☐

decreases



Question 25 3.5 pts

move to light bulb. Use the same simulation

(building on Q 24). Click on the + icone upper left. The peak wavelength is in

☐

green

☐

blue



IR



red



Question 26 3.5 pts

Stay with the light bulb. Multiply peak wavelength and temperature. you get about



1



3000



2900



1500



Question 27 3.5 pts

Playing with the simulation you see that (computations above):



there is no relationship between temperature and peak wavelength



peak wavelength is proportional to temperature



peak wavelength is equal to temperature



peak wavelength is inversely proportional to temperature



Question 28 3.5 pts

The Earth is a black body (approximation) that emits (like us)



I missed the class



IR



visible light



UV



Question 29 3.5 pts

Which spectral type represents stars with spectra that peak at the shortest ultraviolet wavelengths? (UV -> high energy). hint: the hottest stars



B



G



M



K



O



Question 30 3.5 pts

Which scientists explained the correlation with the spectral lines ("QR code") and the temperature of stars?



Meghnad Saha



Annie Jump Cannon



Max Planck



Ludwig Boltzmann

Not saved

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